AMI SYSTEM INFRASTRUCTURE

Empowering the Smart Grid with Advanced Metering Intelligence

DEFINING THE CORE: WHAT IS AMI?

Advanced Metering Infrastructure (AMI) is an integrated system of smart meters, communications networks, and data management systems that enables two-way communication between utilities and customers.

Unlike traditional AMR (Automated Meter Reading), AMI provides a continuous connection, allowing for real-time monitoring and dynamic grid control.

PART 1: HOW TO BUILD AN AMI SYSTEM

THE ARCHITECTURAL PILLARS

Smart Meters

The edge devices that capture sub-second consumption data and events.

Comm Network

The RF Mesh, PLC, or Cellular path used to transport data packets.

MDMS / HES

The software brain that validates, stores, and analyzes meter data.

THE LEAP: AMI 1.0 VS AMI 2.0

Feature	AMI 1.0 (Legacy)	AMI 2.0 (Next-Gen)
Primary Goal	Billing & Simple Reads	Grid Edge Intelligence
Data Frequency	15/60 Minute Intervals	Sub-second Waveform Data
Processing	Centralized (Cloud)	Distributed (At the Meter)
Standards	Proprietary Protocols	Open Standards (Wi-SUN/DLMS)

AMI 2.0: THE DISTRIBUTED EDGE

Edge Computing

Meters now act as IoT sensors, running apps to detect transformer health and load disaggregation directly on the device.

High Speed Backhaul

Integration with 5G and fiber-optic backbones ensures that edge alerts reach the utility in milliseconds.

PART 2: ESTABLISHING COMMUNICATION

THE CONNECTION WORKFLOW

Deployment

Physical install of smart meters across the territory.

Discovery

Meters scan for local nodes or cellular base stations.

Auth

Secure handshake and encryption key exchange with HES.

Sync

Full registration with MDMS and billing systems.

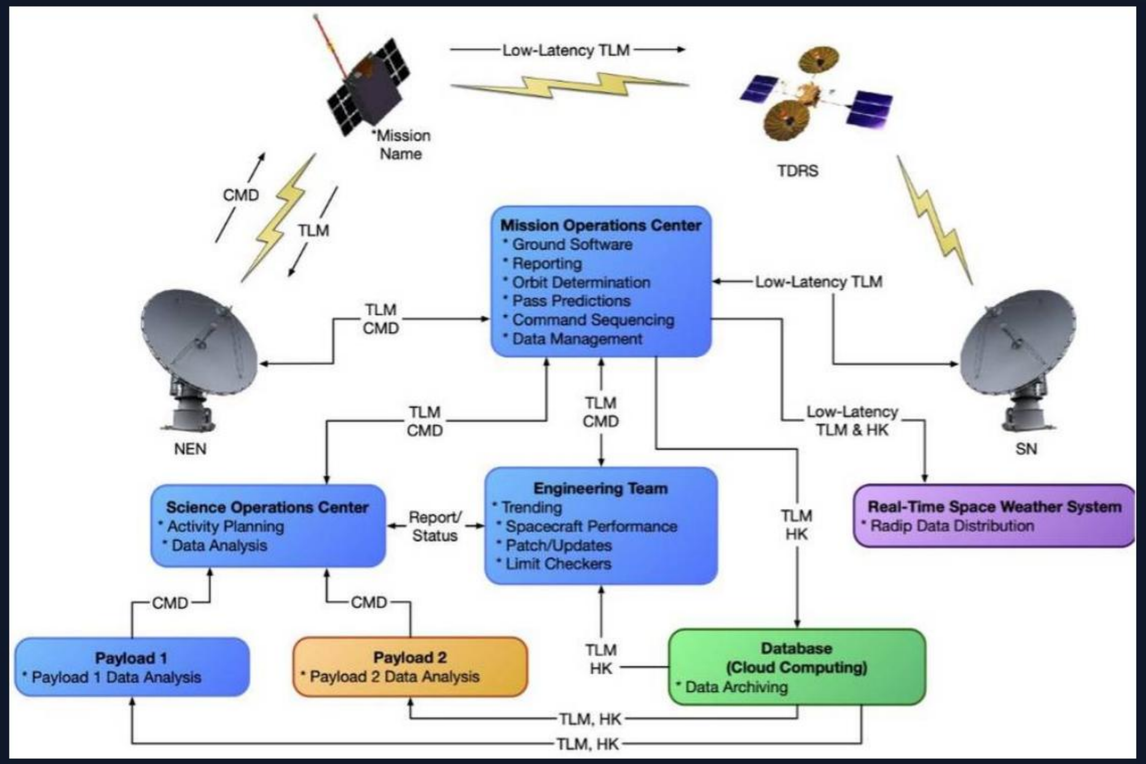
TOPOLOGIES & PATTERNS

RF Mesh Topology

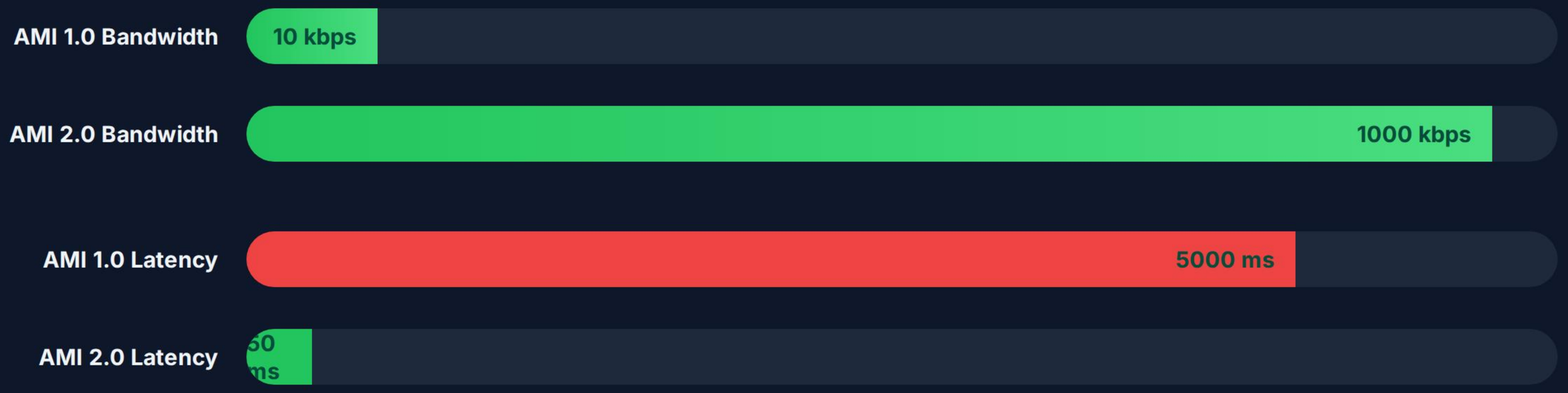
Meters act as repeaters, creating a self-healing web of connectivity. Ideal for high-density urban areas.

Star (Cellular)

Meters talk directly to a tower. High bandwidth and low latency, perfect for AMI 2.0 requirements.



NETWORK PERFORMANCE TRENDS



THE SMART ECOSYSTEM

AMI is the foundational layer for integrating electric vehicles, residential solar, and distributed energy storage into the modern grid.

QUESTIONS?

Advanced Metering Infrastructure Technical Deep-Dive

BUILDING THE INTELLIGENT GRID EDGE

IMAGE SOURCES

https://cdn.prod.website-files.com/67139b4944f3d6b890cda082/68da106118a315a3bc4a1584_freepik__a-closeup-s..._imresizer.jpg

Source: www.workongrid.com

---

https://www.nasa.gov/wp-content/uploads/2021/10/figure11.2-soa2022.jpg?w=985

Source: www.nasa.gov

---

https://images.presentationgo.com/2025/08/futuristic-power-grid-visualization.jpg

Source: www.presentationgo.com